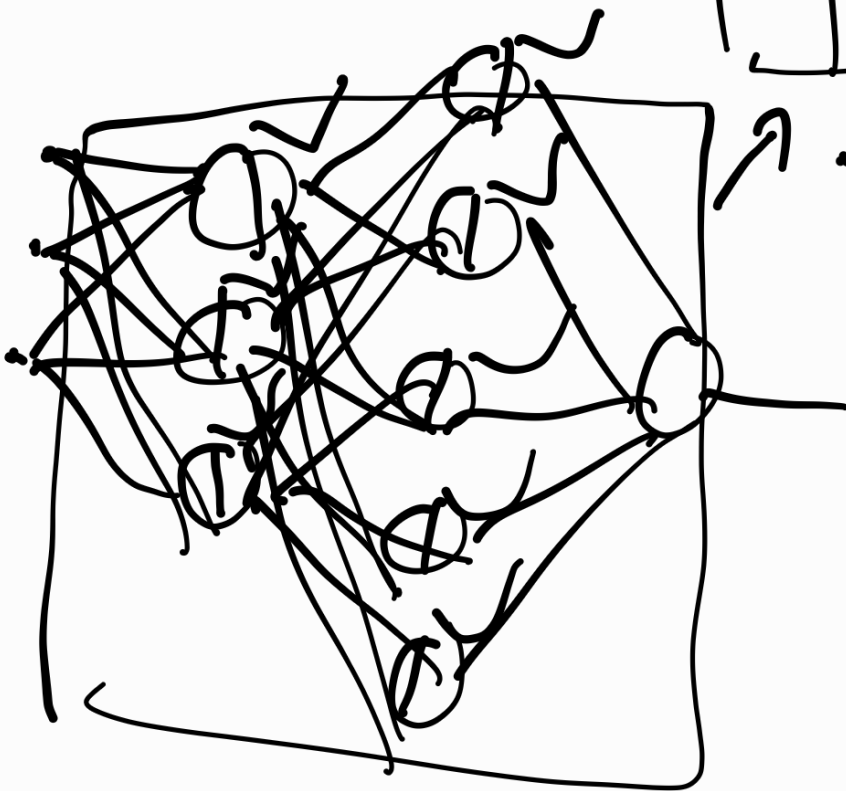
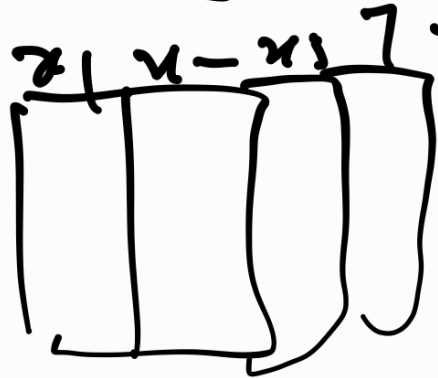
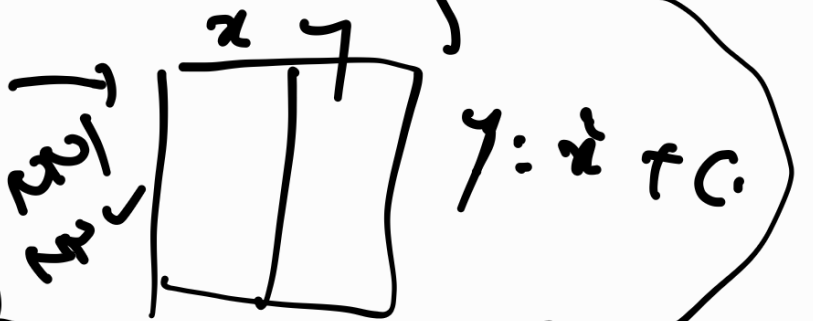
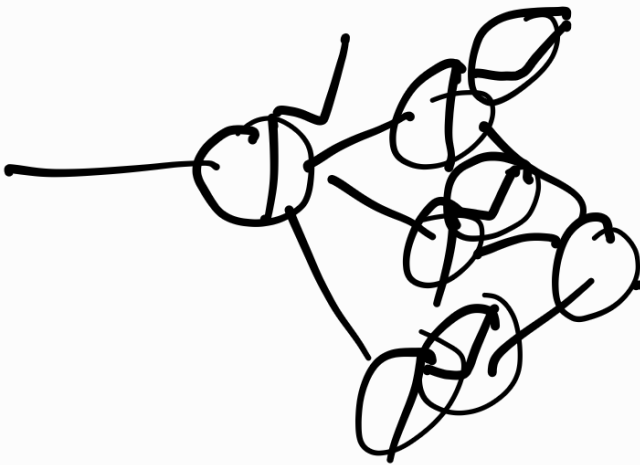
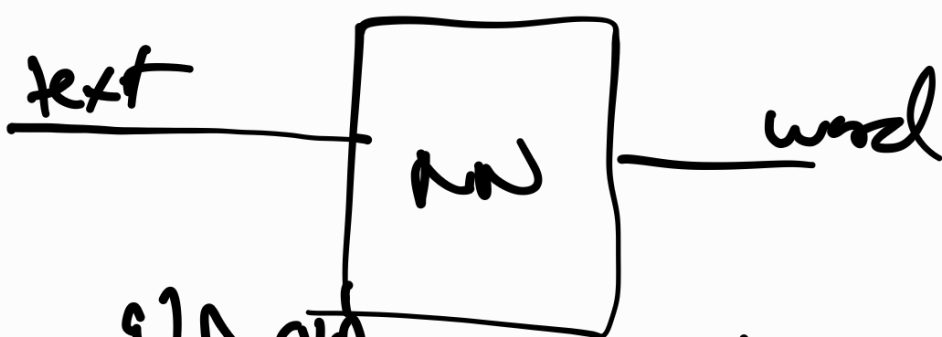


NN



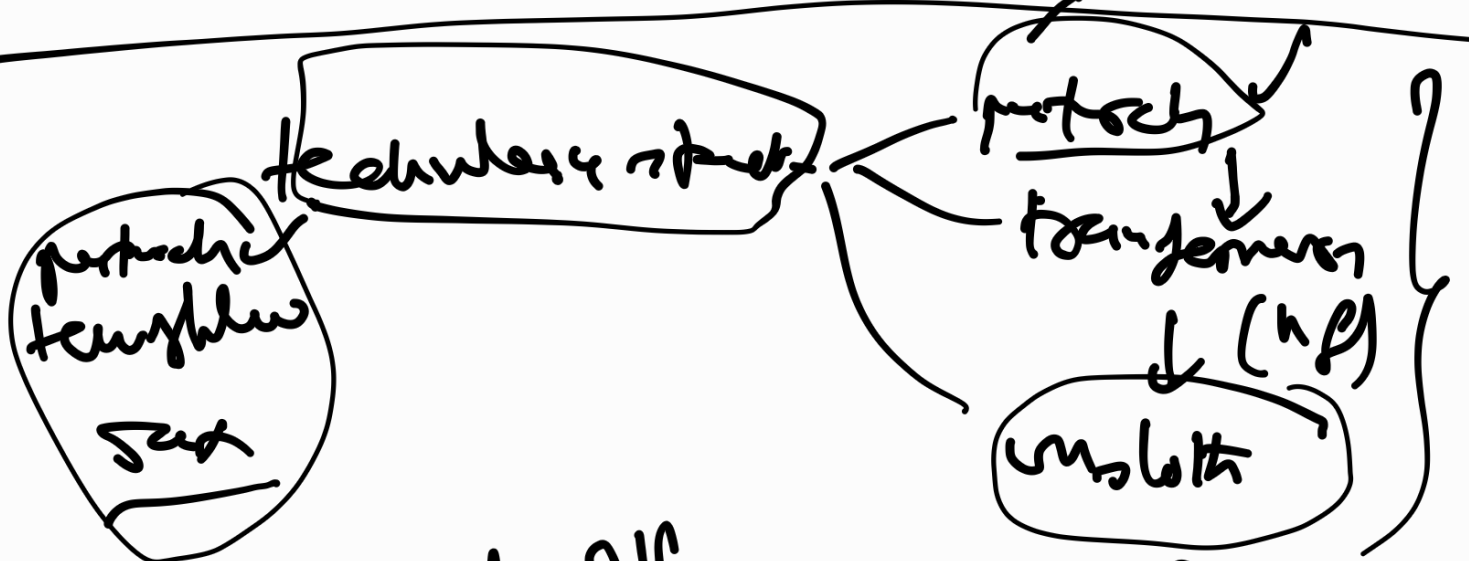
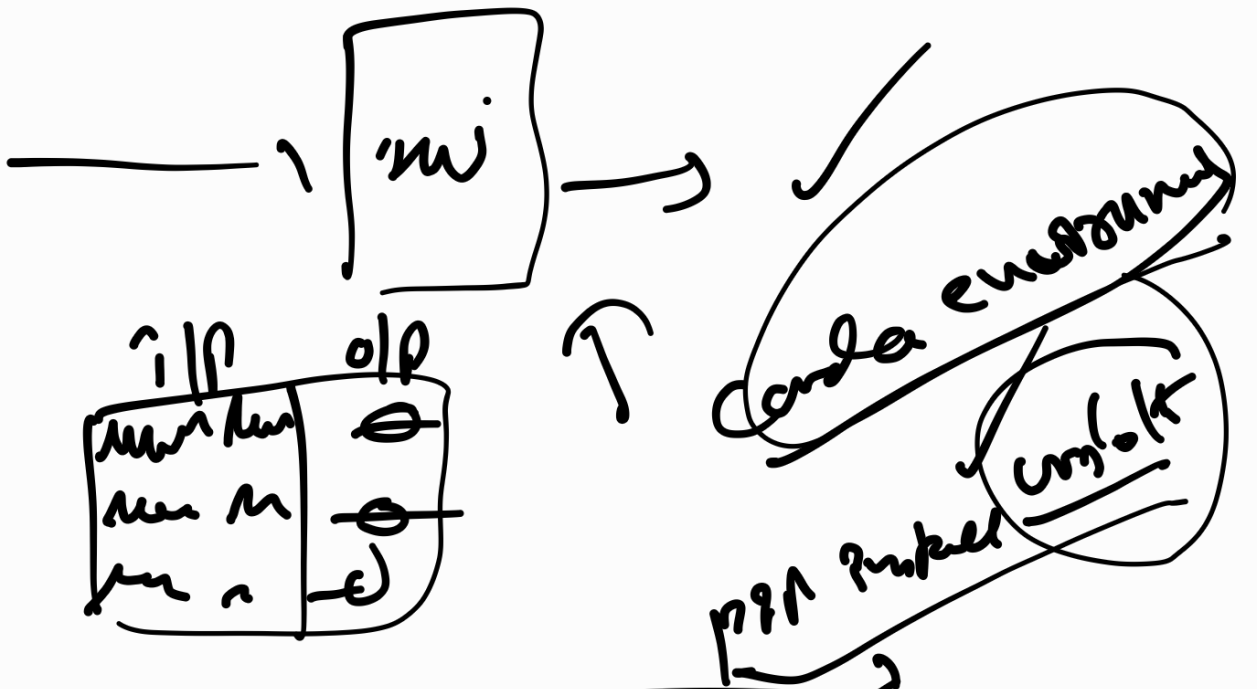
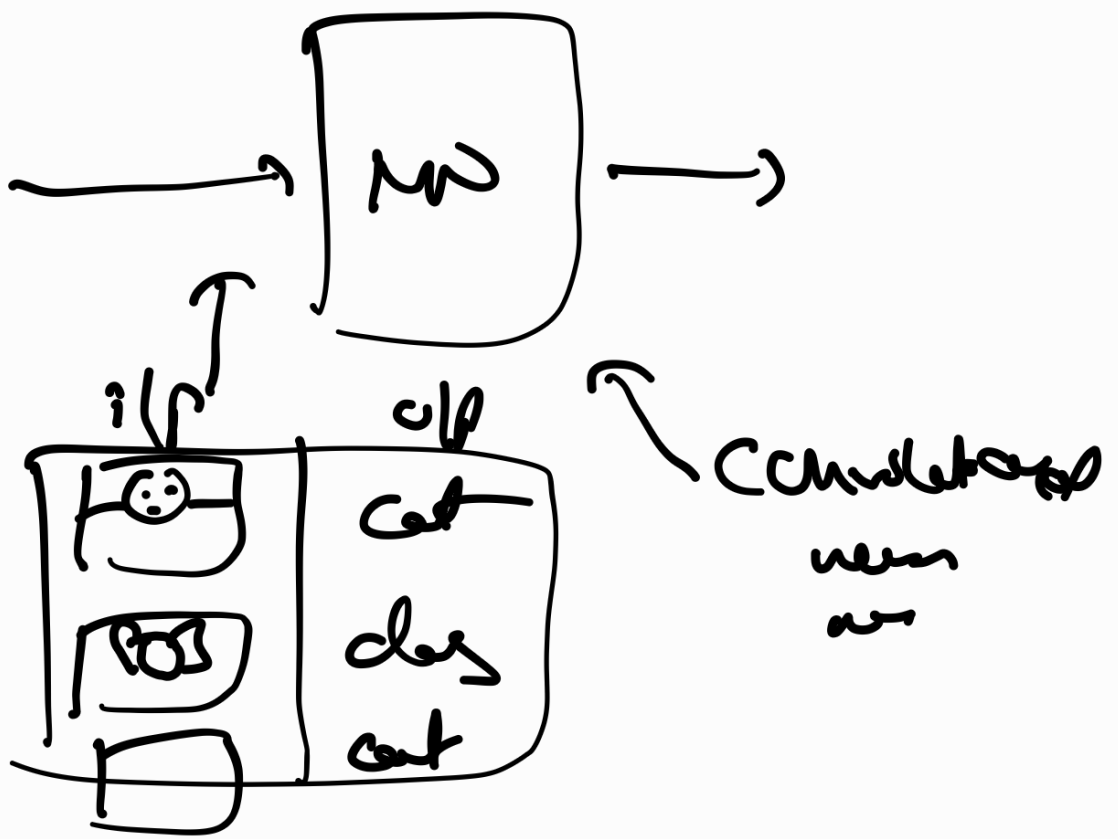
$y = x^2 + 6x + 7$
 x^3
 x^2

NN can learn patterns in unstructured
 data like text, images, voice,
 video, etc.

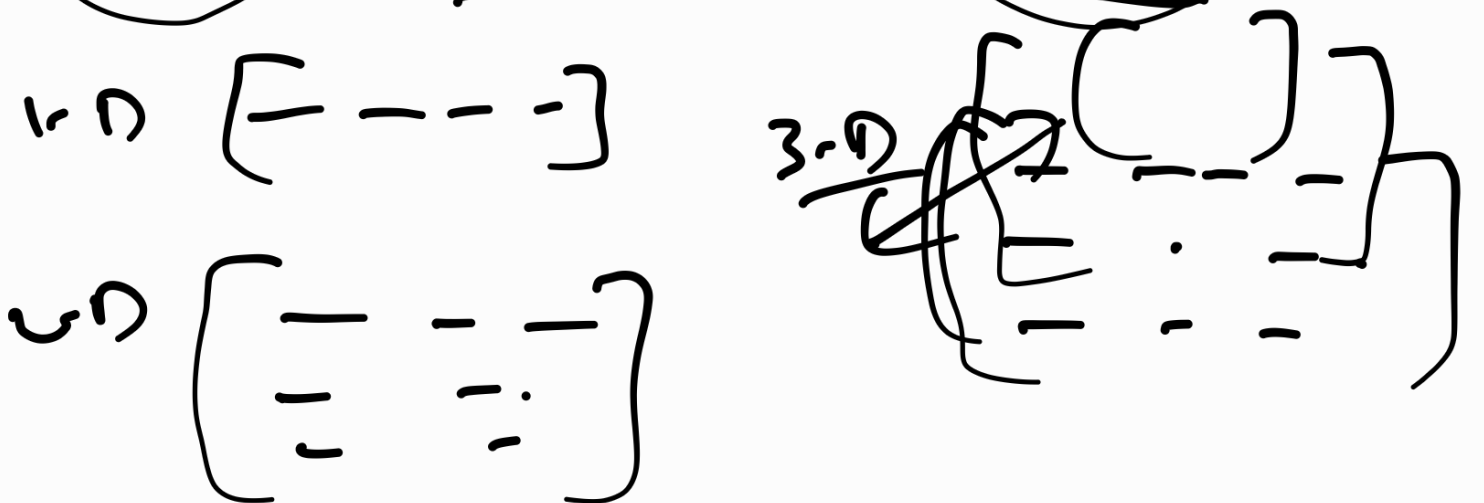


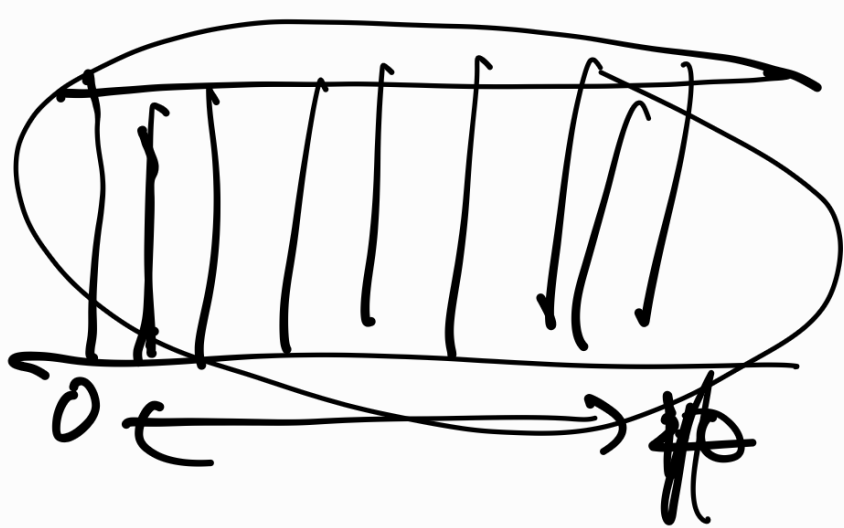
9	1	0	1
i	am	the	best
1	am	the	best
1	am	the	best

transformer
 architecture

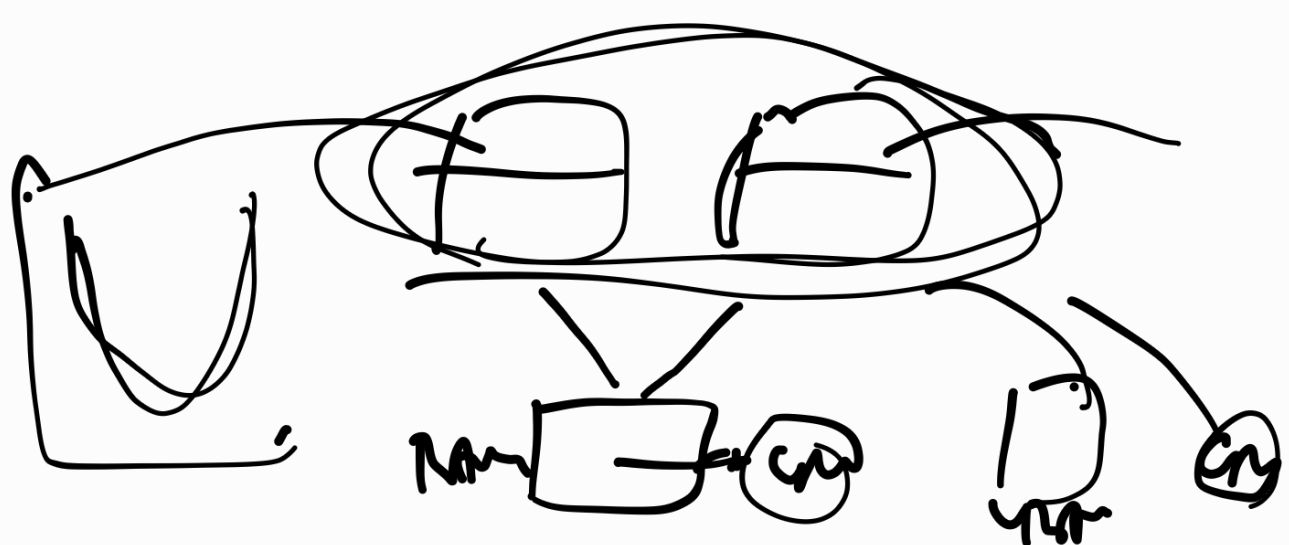
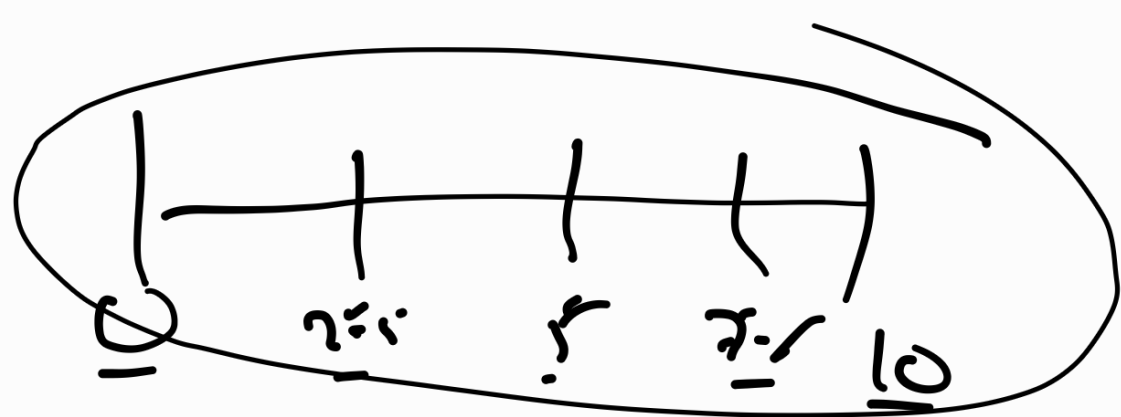
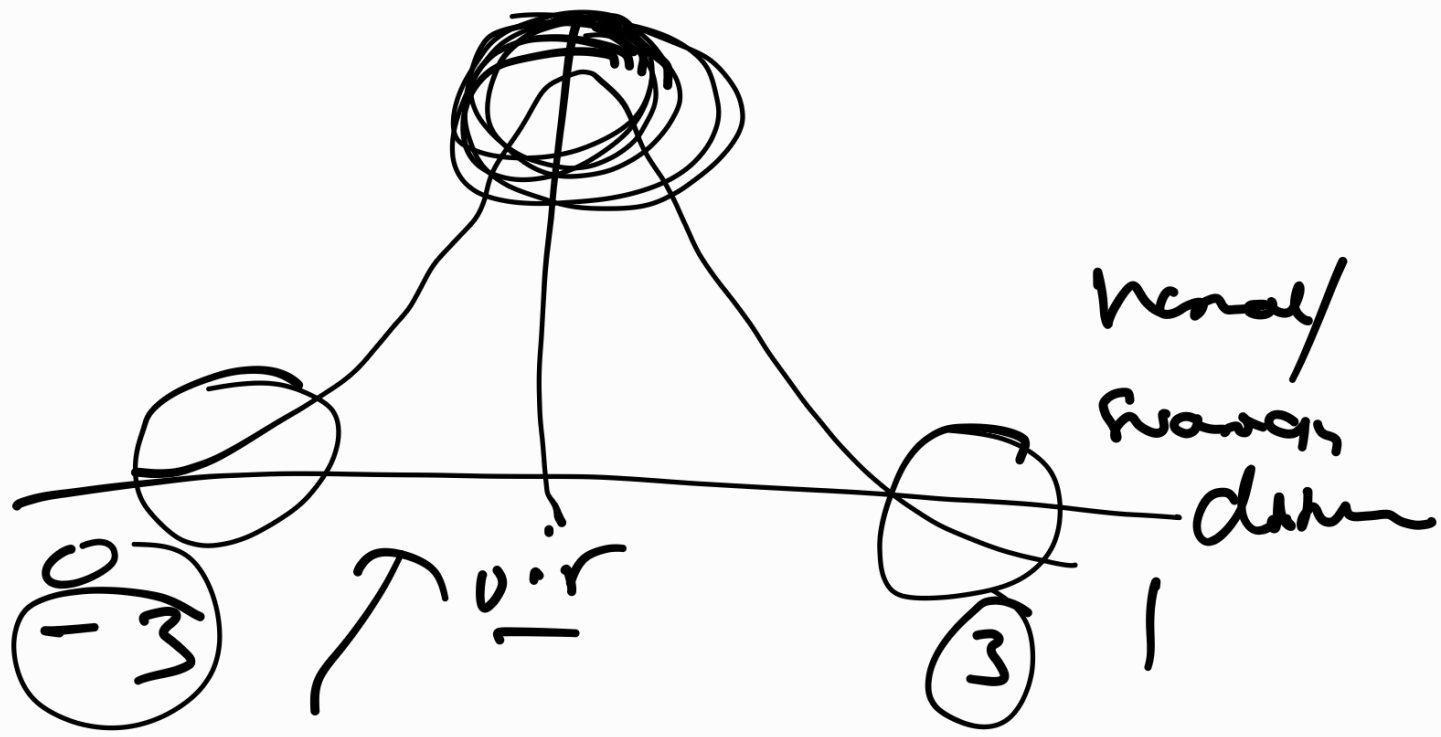


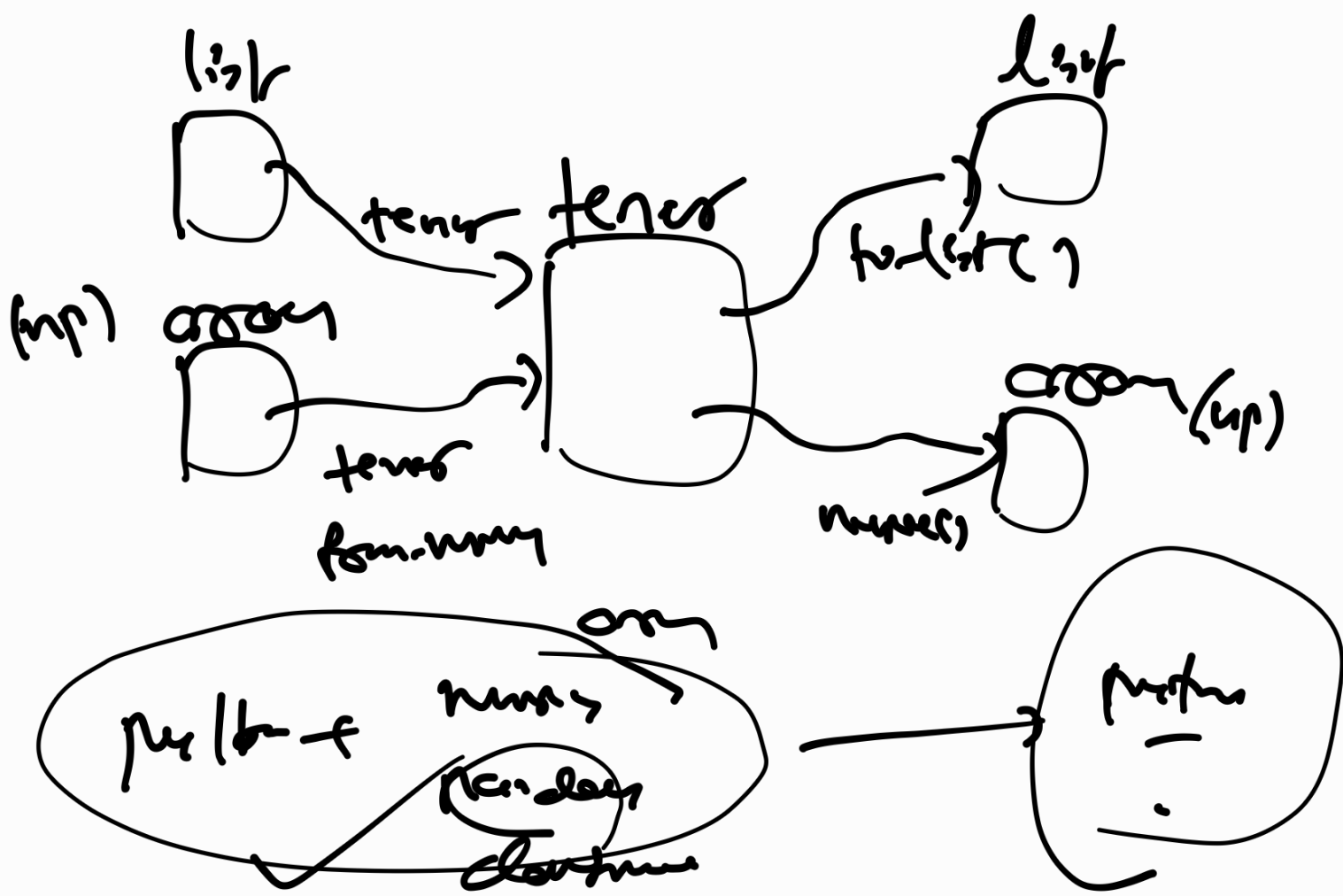
modeller i/p
 & neuron functionally function
 keras multidimensional array



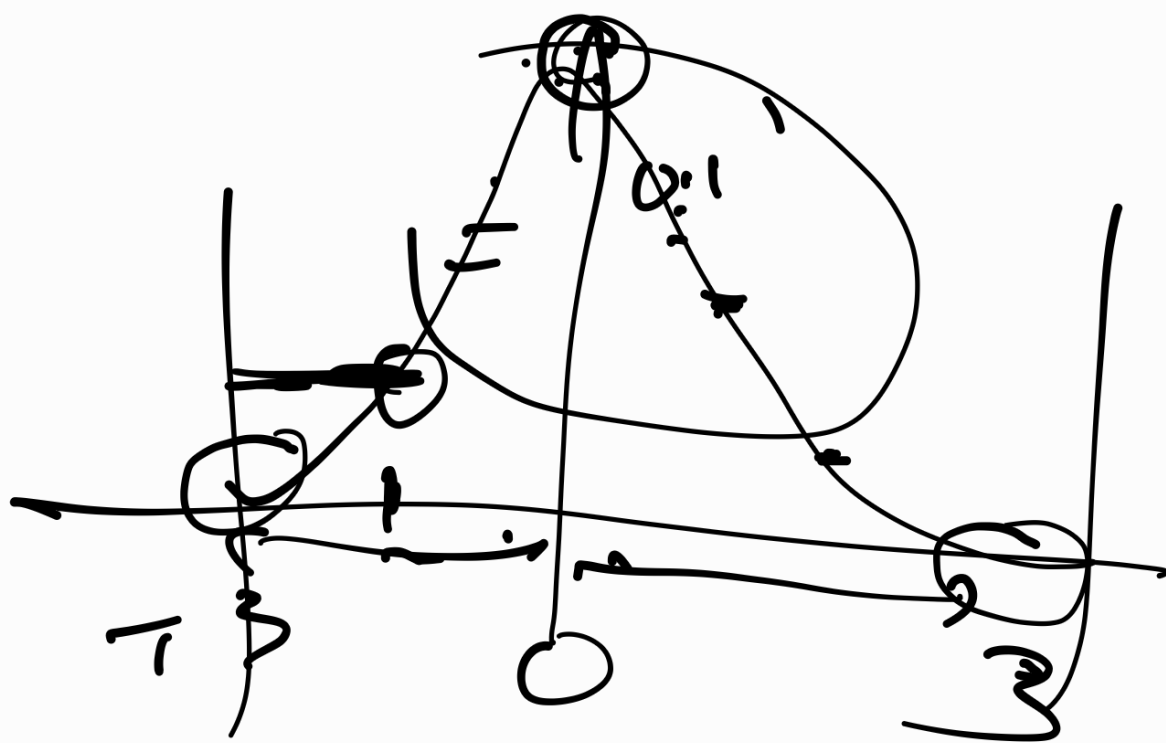


unburn





CUDA → ^{proposed} nuclear singular



$x.view(1, \underline{E})$

[
]
]

detours



$x.view(3, 1)$

{
}
}

$x.view$

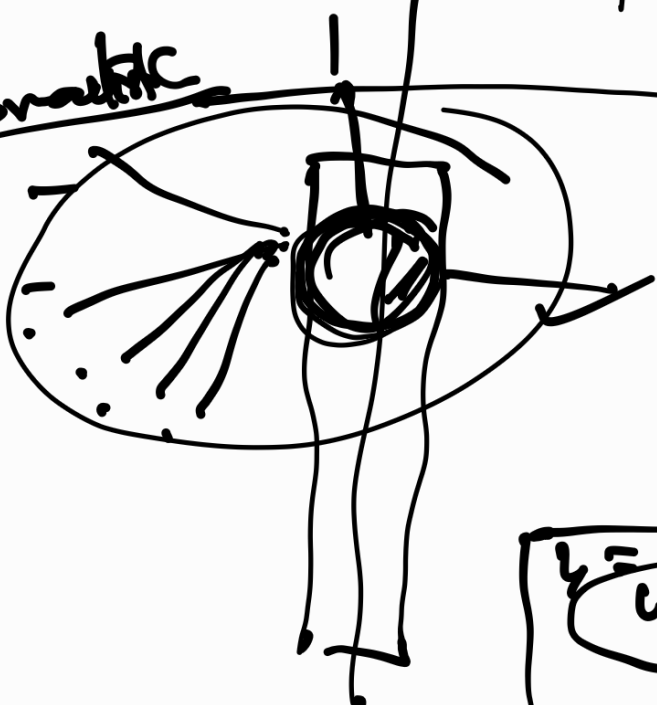


[
(-, -, -)
(-, -, -)
]

[
(-, -, -)
(-, -, -)
]

(x, x, x)

Schematic

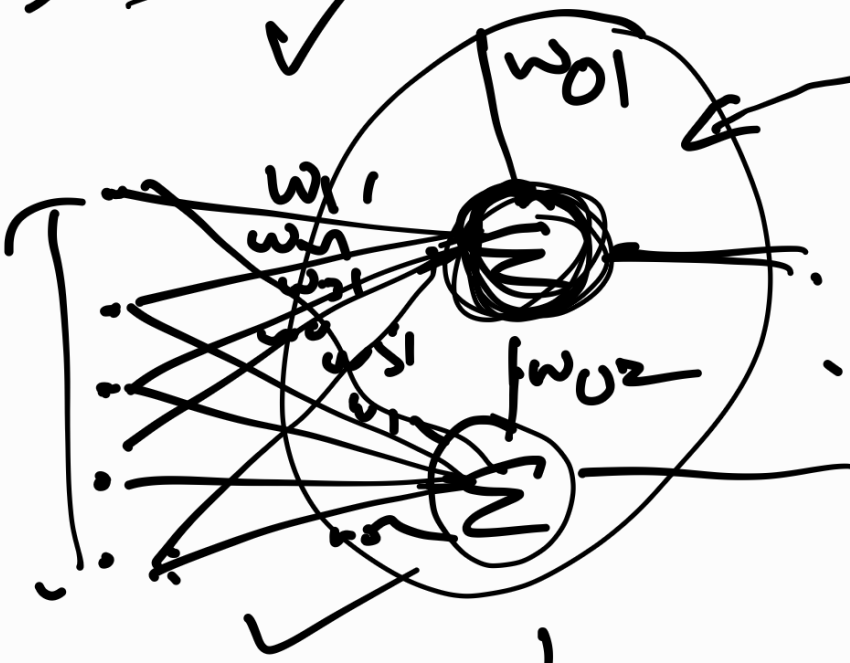


mathematically

$w_1, x_1, w_2, x_2, \dots, w_n, x_n$
+ w_0

$$\underbrace{\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{bmatrix}}_{1\text{-D tensor}} \times \underbrace{\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{bmatrix}}_{2\text{-D tensor}} + w_0 = \dots$$

single axis



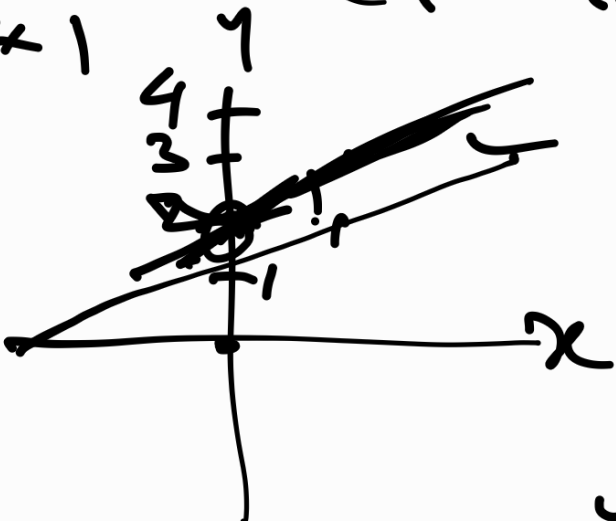
$$\begin{aligned} & w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + w_5 x_5 \\ & + w_0 \end{aligned}$$

mult add

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{bmatrix}_{1 \times 5} \times \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \\ w_{41} & w_{42} \\ w_{51} & w_{52} \end{bmatrix}_{5 \times 2} = \begin{bmatrix} w_{01} & w_{02} \end{bmatrix}_{1 \times 2}$$

$$2 \times 5 = \begin{bmatrix} \quad \end{bmatrix} 1 \times 2$$

W X
or 5x1

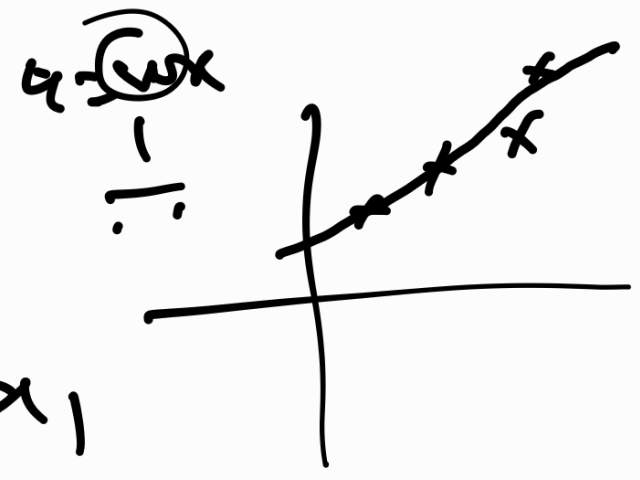
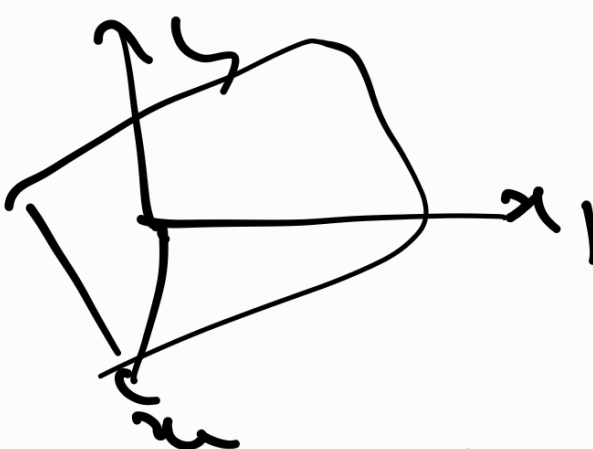


$$\begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} + \begin{bmatrix} w_{01} \\ w_{02} \end{bmatrix}$$

$$y = 2x + 3$$

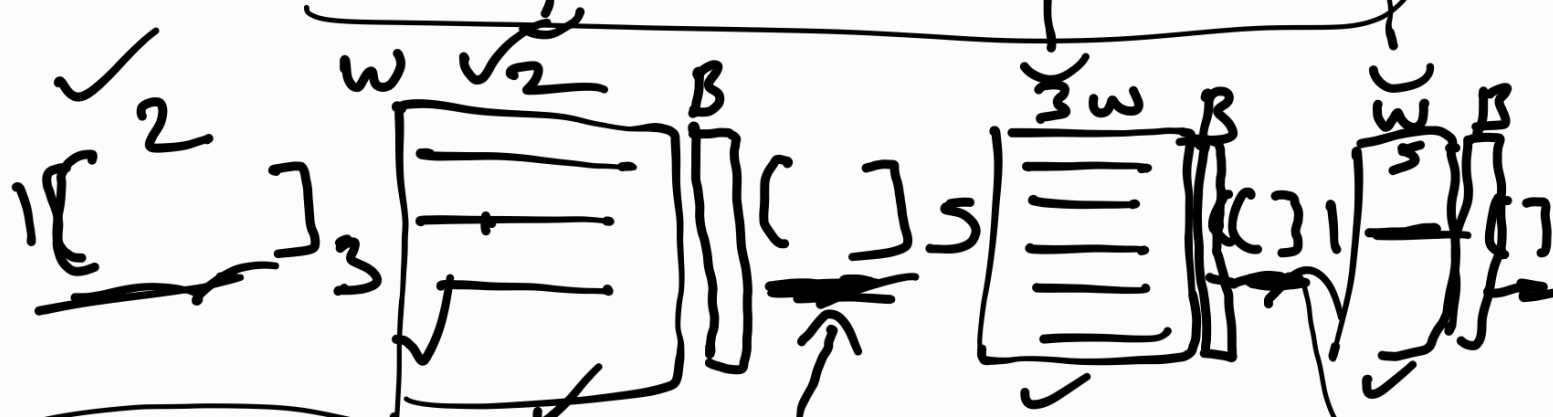
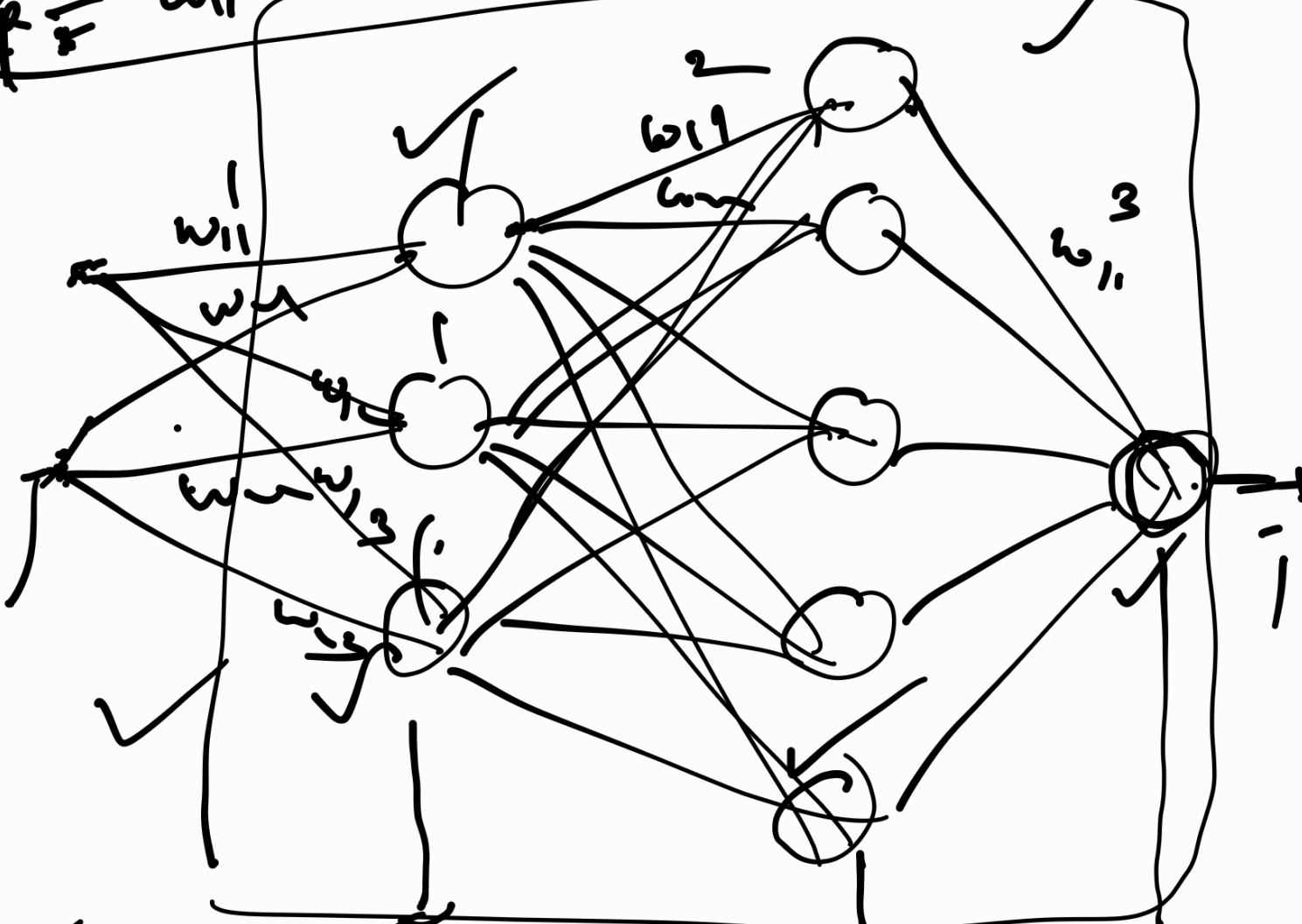
$$y = 1wx + 3w$$

$$y = 2x - 1$$



Multilayer network

$$y = w_1 + \dots$$



$$\begin{aligned} & [1 \times 2] \times [2 \times 3] \\ & (1 \times 3) + (1 \times 3) \end{aligned}$$

$$\begin{aligned} & [1 \times 3] \times [3 \times 5] \\ & (1 \times 5) + (1 \times 5) \end{aligned}$$

$$\begin{aligned} & [1 \times 5] \times [5 \times 1] \\ & (1 \times 1) + (1 \times 1) \end{aligned}$$